Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2022	ENGI3301-WE01

Title: (BEng/MEng)

Structures and Geomatics 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam, submission is time-limited to 4 hours within a 24 hrs window. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the relevant 24-hour window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Question 1

(a) Figure Q1a shows a 2D two-storey frame subjected to 20 kN and 40 kN horizontal loads. The columns of the frame are assumed to have equal cross-sectional areas. Calculate the approximate values of the reactions at C using the portal method of analysis.

[40%]

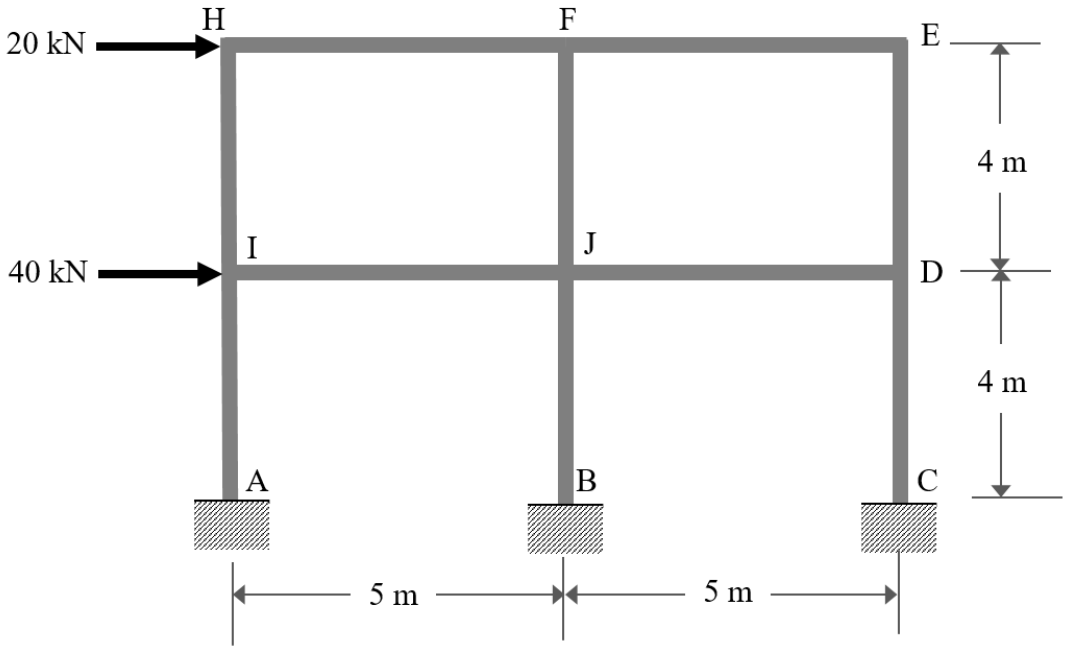


Figure Q1a

(b) Figure Q1b shows a continuous beam comprising three spans. Each span has a length L. The continuous beam is loaded with a uniformly distributed load in span AB of 4 kN/m and a downwards point load of 10 kN in span CD at the middle of the span as shown in the figure.

(i) How many redundancies does this structure possess and why is this significant for structural analysis?

[15%]

(ii) Span AB is a steel beam with Young's modulus of 205 GPa and second moment of area of 67 x 10^6 mm^2. The vertical reaction at A is found to be 6.455 kN and the in-plane rotation at each ends of span AB are found to be

θ_A=-0.0007 rad, θ_B=0.0004 rad

Using the stiffness method, find the length L of span AB.

[45%]

continued

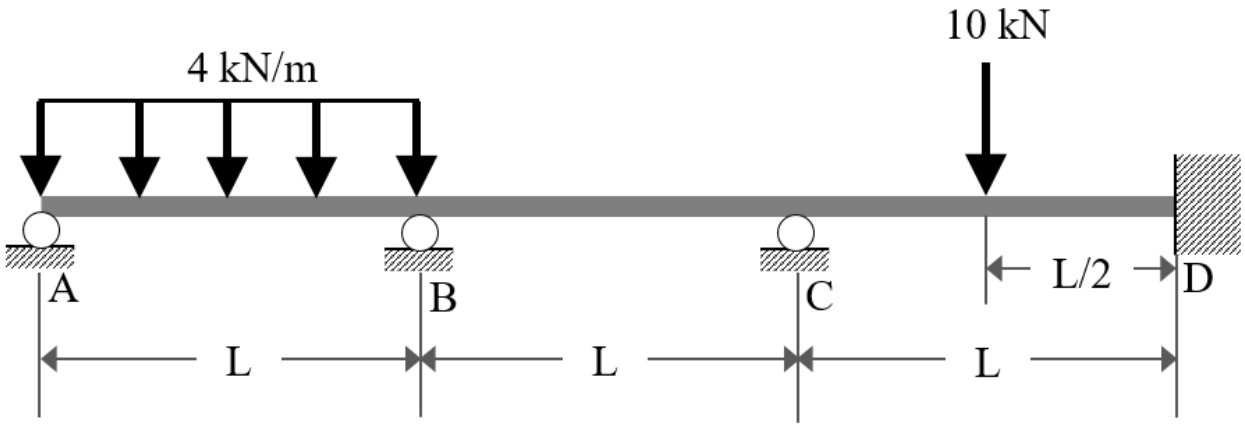


Figure Q1b

continued

Question 2

The pinned-foot portal frame in Figure Q2 carries vertical and horizontal forces. If the plastic moment capacity of columns is $M_p = 300\ kNm$ and the beam $3M_p = 900\ kNm$:

(a) Determine the critical collapse mechanism and the load factor, λ corresponding to the collapse load,

[75%]

(b) Draw the bending moment diagram at collapse. Show the values of the bending moments on the diagram.

[25%]

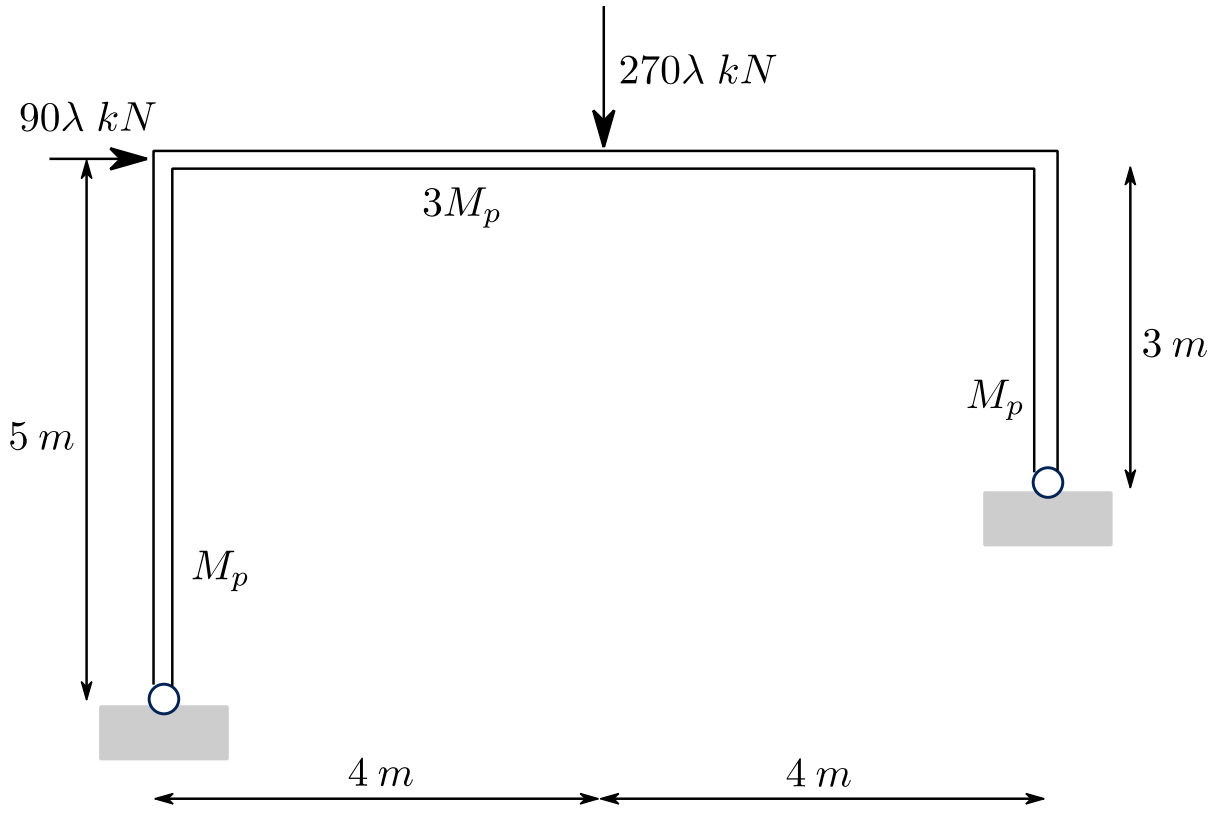


Figure Q2

Question 3

(a) Setting Out

You are working on a large construction project, as a Graduate Civil Engineer. Your task is to set out the curves for the new access road into the main phase of construction works. Your site engineer has provided you with the below information in Table Q3a. Use this data to create tabulated curve setting out information.

[30%]

Table Q3a Setting Out Curve Data

Radius, R	300m
Deflection Angle, Ø	36° 45’ 17’’
Chainage at Q (at the intersection point)	750m

(b) Traverse Survey

Later on in the project, you are now working on the main phase of construction, completing the L-shaped hotel complex. The Senior Site Engineer has completed a traverse survey to form the structural grid for the project. However, you notice the traverse survey has not closed the loop back to Station A. Your task is to use the data collected to calculate the coordinates of Station A then to perform a misclosure check against the known location before ground is broken for the foundations. The Senior Site Engineer has left you an annotated traverse loop (Stations A – H), as shown in Figure Q3 and provided you with a tabulated set of observations and known data in Table Q3b. Comment on the quality of all the survey data and its acceptability.

[40%]

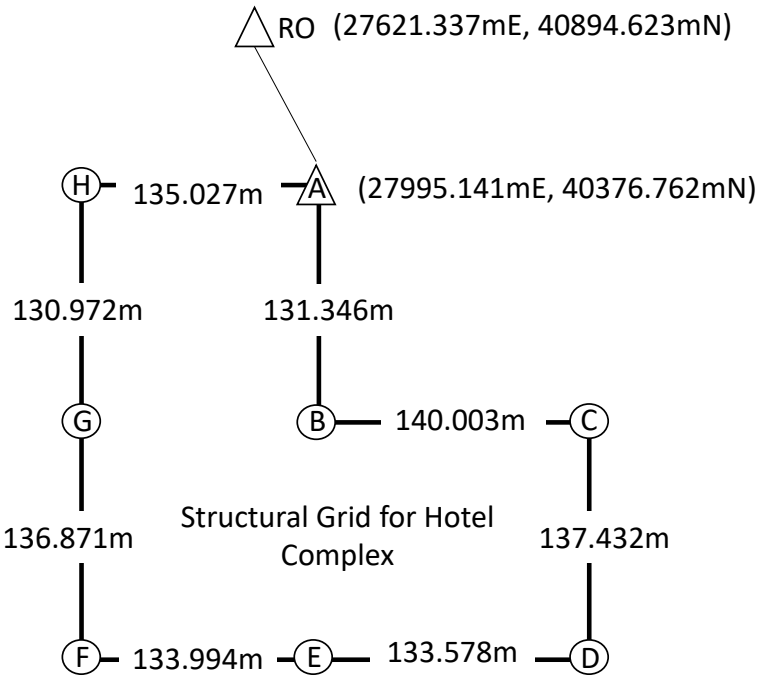


Figure Q3 Annotated Traverse Diagram

continued

Table Q3b Traverse Survey Data

Known Data	
Coordinates of RO	(27621.337mE, 40894.623mN)
Coordinates of A	(27995.141mE, 40376.762mN)
Collected Observations	
Horizontal Clockwise Angle AT-FROM-TO A-H-RO	61° 34’ 06’’
Forward Bearing (H-A)	82° 37’ 04’’
Computed Coordinates	
Coordinates of H	(27861.124mE, 40359.392mN)
Traverse Legs	
A-B	131.346m
B-C	140.003m
C-D	137.432m
D-E	133.578m
E-F	133.994m
F-G	136.871m
G-H	130.972m
H-A	135.027m

(c) Surveying Method Choice

Planning for a second phase of construction on the project to create a spa and golf course next to the hotel complex has begun. The adjoining land is a brownfield site with previous use as an engine depot resulting in the spillage of hydrocarbons, which is visible on the ground surface; the area is largely inaccessible with an uneven terrain. Your task is to create a survey plan for the area, which captures the topographic data, contamination areas, biodiversity of the site (in the form of trees, grass, hard-standing etc.) and any existing structures remaining from the engine depot. You need to make a written argument suggesting two possible surveying techniques that are capable of capturing all the required data. Each technique may capture different types of data and does not have to capture all the required datasets.

[30%]

END